Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document describes how the **Human Development Index Prediction** system can be enhanced to support larger datasets, improved prediction accuracy, and better user experience in the future. It also presents the roadmap for expanding the application's capabilities beyond its current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset for model training	Prediction accuracy depends on the available dataset.	High
2	Single-server Flask deployment	Limited support for multiple concurrent users.	Medium
3	Basic prediction functionality	Does not include advanced data visualization or analytics.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a single Flask server.	Deploy the application on cloud platforms with auto-scaling support.
2	Dataset	Uses the existing HDI dataset for prediction.	Expand the dataset with recent global development statistics for improved accuracy.
3	Performance	Prediction is generated using a single trained model.	Optimize the model and improve response time using efficient deployment techniques.
4	User Interface	Basic web interface for prediction.	Develop a more interactive, responsive, and user-friendly interface with graphical visualizations.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase-2	Integrate updated HDI datasets from trusted global sources.	Future Release	Improves prediction accuracy and data reliability.
Phase-3	Add interactive charts and country-wise HDI comparison.	Future Release	Better visualization and easier analysis of development trends.
Phase-4	Deploy as a cloud-based application with enhanced scalability and security.	Future Release	Supports more users and provides faster, secure access from anywhere.